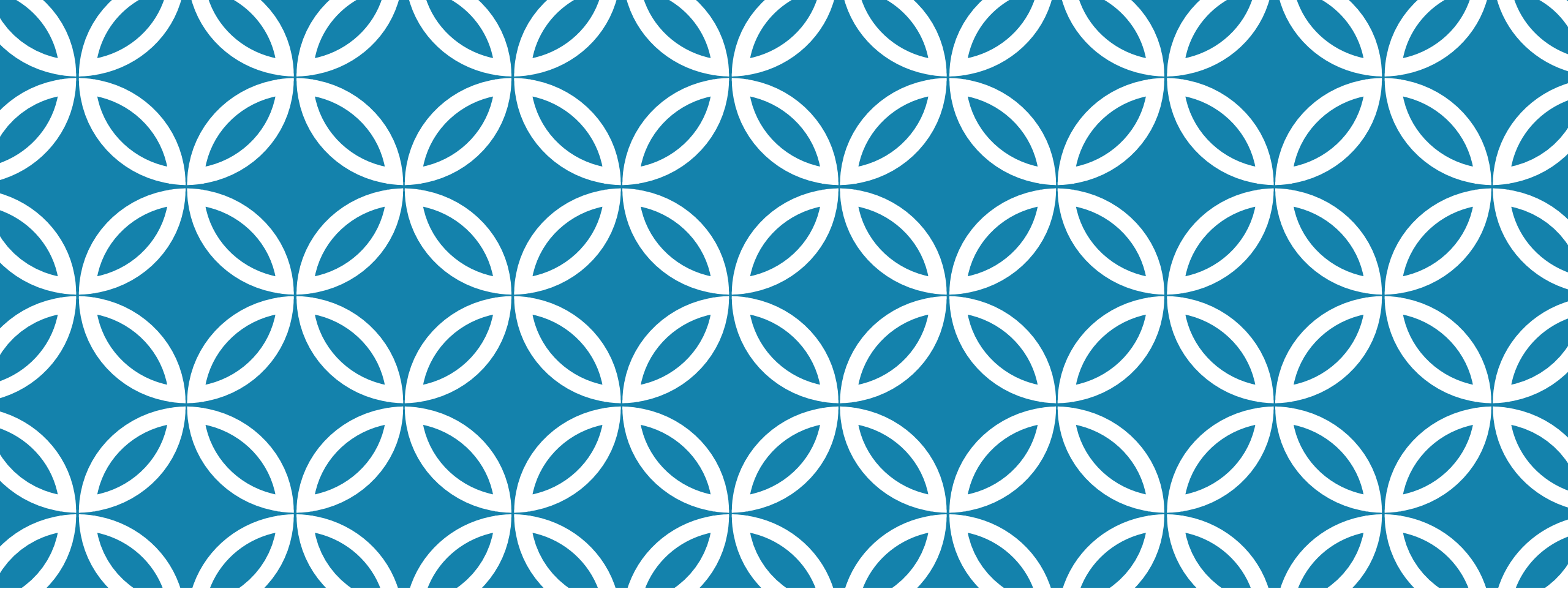




METHOD OF BUILDING ESTIMATE

Lecture IV

GENERAL ITEMS OF WORK FOR BUILDING ESTIMATE

The general items of work involved in the estimation for a building are listed below:

- Earthwork in excavation for foundation trenches.
- Earthwork infilling.
- Brick flat soling.
- Cement or lime concrete in foundation.
- Masonry work in foundation (brickwork or stone).
- Damp Proof Course(D.P.C.).
- Masonry work in superstructure walls.
- 10 cm thick brickwork.
- R.C.C. or R.B. work.
- Steel work.
- Centering and shuttering (form work).
- Lime terracing (lime concrete in roof).
- Woodwork for doors and windows.
- Mild Steel (M.S.) clamps or holdfast.
- Flooring.
- Plastering and Painting items.
- White washing/ Colour washing/ Distempering.
- Painting for woodwork or miscellaneous items.

PROCESS OF CALCULATION

a

EARTHWORK IN EXCAVATION FOR FOUNDATION TRENCHES

Earth is excavated for foundation trenches to the exact width and depth of foundation with vertical sides and bottom levelled up both longitudinally and transversely. The quantity of earthwork is measured in cubic metre. (Earthwork for steps is also included.)

b

EARTHWORK INFILLING

This consists of two parts: a) Foundation trenches and, b) Plinth infilling. Usually excavated earth is used for filling. Considering quality of earth, plinth filling might also be done using sand in which case, plinth filling work shall be separated from foundation filling.

- a. **Foundation filling** – Considering the low rate of earth filing, usually the quantity is taken to be $1/5^{\text{th}}$ of excavated earth volume. Though for more appropriate results, it is calculated by: “Volume of work in excavation – volume of work in foundation”. Unit is cubic metre.
- b. **Plinth filling** – The internal length and breadth, of a room or verandah are considered and multiplied by the depth of filling, which is measured from ground level up to the bottom of floor structure. Unit is cubic metre.

PROCESS OF CALCULATION

c

BRICK FLAT SOLING

As cement concrete is usually used in foundation now a days, which does not require ramming, a layer of brick soling or lean concrete (in poor proportion) is provided. Soling is measured in **sqm**. by multiplying length and breadth of the trench. In case of lean concrete, the volume of concrete is given in cubic metre by multiplying thickness of concrete with length and breadth.

d

CEMENT OR LIME CONCRETE IN FOUNDATION

The volume is calculated in cubic metre by multiplying length, breadth and, depth of concrete. Length and breadth remain same as that of earthwork, provided all the sections of walls in the building are same. (In case of different sections, the length varies).

e

MASONRY WORK IN FOUNDATION OR PLINTH

The volumes of different offsets or layers and plinth wall are calculated one by one, in cubic metre by multiplying their individual length, breadth and height (thickness). No deduction is required as there is no opening in wall up to plinth. This can be brick work or stone work.

PROCESS OF CALCULATION

f

DAMP PROOF COURSE (D.P.C.)

This is a layer of cement concrete mixture in proportion of 1:2:4 mixed with water proofing compound laid in between the plinth and superstructure walls to prevent the rise of water by capillary action from the ground. Usually paired with layer of hot bitumen, or an array of different water proof substances like cement mortar. Measured in sqm by calculating length and breadth of superstructure walls. Deduction in area is made from door, verandah or any other such openings, whose bottom starts from plinth level.

g

MASONRY WORK IN SUPERSTRUCTURE WALLS

For walls of one brick thickness, volume is calculated in cubic metre by multiplying length, breadth and, height. Standard deductions for openings are made from the total volume of the wall (doors, windows, lintels), as per IS:1200 standards. Wall shelves and cupboards thickness are worked out separately added if required. For multi-storeyed buildings, volume is to be worked out separately for each storey.

PROCESS OF CALCULATION



h

10 CM THICK BRICKWORK

10 cm thick or when the length of the brick is laid parallel to the wall, the quantity of brickwork is measured in sqm by multiplying length and height. Any brickwork more than 10 cms or half brick (in case columns are provided to support 10 cm walls), the calculation is to be done separately. Same goes for honeycomb brickwork or brick jallis (unit sqm , no deductions for holes in jallis)



i

R.C.C. or R.B. (Reinforced Brickwork)

Reinforced cement concrete is used in slabs, lintels, columns etc. The quantity is measured in cubic metre by multiplying length, breadth, and height including reinforcement. Shuttering is classified separately and calculated independently as per IS recommendations of formwork. If reinforcement and shuttering is considered as one consolidated item, this should be specified in the description and calculated as R.B. work. In cubic metre.



j

CENTERING AND SHUTTERING (FORMWORK)

The cost of formwork is about 30% of cement concrete. Unless mentioned otherwise, formwork is measured separately in sqm by multiplying length and breadth. Generally the surface area in contact with concrete is measured.

PROCESS OF CALCULATION

k

STEEL WORK

Reinforcement or steel is measured by weight in quintal including binding wire. No separate payment is made for binding wire. For estimation purpose, the weight of reinforcement may be calculated from the percentage volume in R.C.C. such as, 0.8% or 1% reinforcement for different members.

The percentage method is the short-cut method for estimating the quantity of reinforcement. In practice, it is calculated from bar bending schedule and paid accordingly.

LIME TERRACING

l

This is measured in sqm multiplying length and breadth. The average consolidated thickness is described. The clear surface area between parapets is measured. Rounded edges are separately noted in description.

PROCESS OF CALCULATION



m

WOODWORK FOR DOORS AND WINDOWS

- a. Door and window frame: The quantity is worked out in cubic metre. The total length of a frame is found first and then is multiplied by two sides of its cross-section. Doors are now provided without bottom sill. So the total length of doors is considered two vertical members + one top member. Size of doors or windows are guided by size of openings. So, length of one vertical member is the height of the opening whereas, the measurement of top member gives the width. Insertions in between verticals and horizontals are considered. Window frame may be provided with bottom sill and intermediate member in between. Then the total length becomes 2 verticals + 3 horizontals.
- b. Door and window leaves or shutters: Shutters are measured in sqm by multiplying height with breadth (or thickness) mentioned in the description. The depth of rebates is also added to it. For double shutters, the overlapping at the centre in between the leaves is not accounted. The design and class of wood is mentioned in the description column.

PROCESS OF CALCULATION

Hinges, tower bolt, sliding bolt, door handles all are described separately, and paid per piece after counting the same. Recommendations given by PWD are to be followed strictly while making the estimate.

MILD STEEL CLAMP OR HOLDFAST

This member is specified in the estimate as numbers. Generally 6 nos. of clamps @ 3nos. For each vertical member in doors are provided. For windows 4 nos. @2nos. For each member is considered. Size and type is stated in the description. The payment for this item may also be done in kg. after weighing one such member. The estimate is included with the fixing cost by cement concrete (1:2:4). Cost of cement may be paid together or separately.

FLOORING

a. Flooring for Ground floor :- The floor consists of two parts:

- First, bottom or bed of floor usually provided with 7.5cm thick lime concrete (1.5:2:7) or cement concrete (1:3:6) over a brick flat soling.

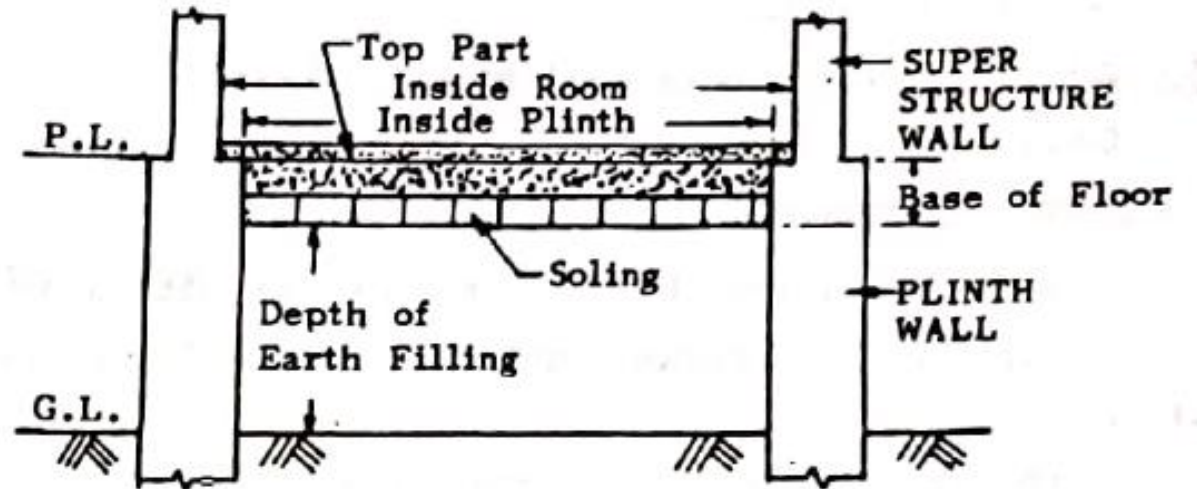
PROCESS OF CALCULATION

- Second, the top part above the base which may be of different types, such as 20mm to 35mm thick artificial stone flooring finished with 3mm to 6mm thick topping with colouring pigment, highly polished and grinding.

The quantity for base 7.5cm thick concrete is calculated in sqm stating the thickness by multiplying the *inside dimensions of plinth to plinth walls*.

Thus, the quantity of top of flooring is calculated also in sqm stating the thickness by multiplying the *inside dimensions of a room*.

So, the area of top of flooring shall be **more** than the area of base concrete.



PROCESS OF CALCULATION

b. 1st floor or subsequent top floors :- The flooring is done on the roof of the lower storey. So, the flooring requires only the top part. To make a finished top of the roof surface a layer of 25mm to 40mm thick cement concrete (1:2:4) with 5mm stone chips is provided in the case of artificial stone flooring.

PLASTERING AND PAINTING ITEMS

p It is measured in sqm. Surface area of walls are taken. No deductions are made for ends of posts, steps or opening not exceeding 0.5 sqm. For openings between 0.5-3 sq m, when both faces are plastered (like internal wall), deductions are made for one face and jambs /soffits are not added. For openings more than 3 sq m, both faces are deducted but jambs, soffits and sills are measured. Plastering bands up to 30 cm are measured in r m (running meter). Dimensions of ceiling are taken before plastering walls for calculation.

WHITE WASHING/ COLOUR WASHING/ DISTEMPERING.

q The number of coats are stated in description. Same deductions as above, corrugated surfaces are measured flat.

PAINTING FOR WOODWORK OR MISCELLANEOUS ITEMS.

r Measured flat in sqm. for painting up to 10 cm in width, it is measured in r m. for uneven surfaces, it is adjusted using following table.

For item nos. (i) to (v) measurement includes chowkat or frame, Edges, Chocks, Cleats etc.

Description of Work		How measured	Multiplying Factor (for each side)
i.	Panelled or framed and braced or ledged and battened or ledged, battened & braced joinery.	Measured flat	1.30
ii.	Flush joinery ...	Measured flat	1.20
iii.	Fully glazed or gauzed joinery ...	Measured flat	0.80
iv.	Partly panelled and partly glazed or gauzed joinery ...	Measured flat	1
v.	Fully venetianed or louvred joinery ...	Measured flat	1.80
vi.	Weather boarding ...	Measured flat; supporting frame work shall not be measured separately	1.20
vii.	Boarding with cover fillets & match boarding	Measured flat	1.05
viii.	Steel roller shutters ...	Measured flat (size of opening) overall; jamb guides, bottom rails etc. shall be included in the item. (cover shall be measured separately)	1.10
ix.	Plain sheet steel doors and windows ...	Measured flat including frame, edges	1.10
x.	Fully glazed steel doors and windows ...	Measured flat including frame, edges	0.50
xi.	Partly panelled and partly glazed or gauzed steel doors. ...	Measured flat including frame, edges	0.80
xii.	Collapsible gate ...	Measured flat(size of opening)	1.50(all over)
xiii.	Trellis (or jaffri) work one-way or two-way	Measured flat overall; no deduction shall be made for open spaces; supporting members shall not be measured separately	2(for painting all over)
xiv.	Guard bars, balustrades, gates, gratings, grills, expanded metal and railings.	Measured flat overall; -do-do-	1 -do-

PROCESS OF CALCULATION



WATER AND SEWER LINES

Measured in r m. measurements taken along central line. Surface drains are measured in r m.

Manholes, sanitary fittings, door handles – measured in item numbers.

IRON WORK

Steel work in trusses – quintal

Cast iron brackets, manhole covers – in kg



Bolts including nuts and washers – in kg

Collapsible gate in sq m (full description of bottom runners, pulleys, locking arrangement given.

Steel doors and windows – sq m

Wire fencing – rm

UNITS OF DIMENSIONS FOR MATERIALS & WORKS

S.No	NAME OF MATERIALS AND WORKS	DIMENSIONS IN METRIC SYSTEMS
1.	Bricks, stone blocks etc	All dimensions in cm.
2.	Aggregates, ballast, grit, sand etc.	Size in mm.
3.	A.C.Sheets, tiles, slates, wall board, glass panes etc	Length and breadth in cm. or m. Thickness in mm.
4.	Timber	Length in m and cross sectional dimensions in cm or mm.
5.	Mild steel bars	Length in m and diameter in mm.
6.	Rolled steel sections	Length in m and section in mm.

UNITS OF DIMENSIONS FOR MATERIALS & WORKS

S.No	NAME OF MATERIALS AND WORKS	DIMENSIONS IN METRIC SYSTEMS
7.	Masonry (brickwork, stone masonry etc)	Length and height in m. Thickness or breadth in cm.
8.	Concrete work (cement or lime concrete, R.C.C. work, concrete flooring etc.)	Length and breadth in m. Thickness in cm.
9.	Door, windows etc.	Height and breadth in m or cm.
10.	White washing, colour washing, distempering, painting etc.	Length and breadth or height in m.
11.	Hardware articles.	Size of the articles in cm or mm.

PRINCIPLE UNITS FOR VARIOUS ITEMS OF WORKS

- a. Mass, voluminous and thick works shall be taken in cubic unit or volume (viz. cubic metre)
- b. Thin, shallow and surface work shall be taken in square unit or in area (viz. square metre)
- c. Long and thin work shall be taken in linear or running units, and linear measurements shall be taken (viz. running metre)
- d. Piece work, job work etc. shall be taken in number.

LIMITS OF MEASUREMENT AND DEGREE OF ACCURACY IN ESTIMATING

- During preparation of an estimate of main head smaller dimensions in its various sub-heads should not be neglected as those affect the total quantity of main head. No approximation should be made after omitting fractional dimensions either directly from the drawings or from those determined from the plan.

LIMITS OF MEASUREMENT ACCORDING TO SP : 27-1984

- a. Dimensions shall be measured to the nearest 0.01m except for (i) thickness of slab or R.C.slab which shall be measured to nearest 0.005m; (ii) woodwork to nearest 0.002m; (iii) steelwork to nearest 0.001m; reinforcement to nearest 0.005m.
- b. Areas shall be worked out to the nearest 0.01 sqm. For steelwork areas excluding cross sectional measurements shall be worked out to nearest 0.001 sqm.
- c. Cubic contents shall be worked out to nearest 0.01 cu.m. except for woodwork shall be worked out to nearest 0.001 cu.m.
- d. Weights shall be worked out to nearest 1kg.

LIMITS OF MEASUREMENT AND DEGREE OF ACCURACY IN ESTIMATING

RATES

- The degree of accuracy in calculations depends upon the rate of the item of work.
- Thus, where the rates are per hundred (per %) or per thousand (per ‰) units, greater accuracy is not required.
- But where the rates are per metre (per m) or per running metre (per rm), per square metre (per sqm), per cubic metre (per cu m) calculations should be carried out up to two places of decimal for greater accuracy at higher rate.

THANK YOU